Web Data Integration
Project Outline

Pharmaceutical Drug Integration: Linking US, EU and Canadian Approval Data

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1 Use Case

This project aims to build a unified view of approved pharmaceutical drugs in the United States, the European Union and Canada, enabling researchers and analysts to query regulatory status, approval history, and product attributes across these markets from a single integrated data set. To achieve this, we combine data from three complementary sources. Drugs@FDA is the FDA's official database of US-approved drugs. EMA Medicines is the European Medicines Agency's register of EU-authorised medicines. The Health Canada Drug Product Database (DPD) contains current and historical drug products from Health Canada. The central entity is a medicine, represented by its brand name and active substance or combination of active substances. Each integrated record includes the licence holder, regulatory status, product class, ATC code, dosage form, route, strengths, active substance count, regional regulatory dates and generic, biosimilar or orphan flags.

2 Requirements Suitability

2.1 Schema and basic profile

Table 1: Overview of the input datasets

Dataset	Source*	Format	Rows	Entities	Attr.	List of attributes**
Drugs@FDA	U.S. FDA	TXT/TSV	51,839	8,175	56	ApplNo, ProductNo, Form, Strength, ReferenceDrug, DrugName, ActiveIngredient, ApplPublicNotes**, SponsorName, SubmissionType, SubmissionStatus, SubmissionsPublicNotes**, ReviewPriority**, MarketingStatusID, TECode, …
EMA Med.	EMA	XLSX/CSV	2,746	2,620	39	Name of medicine, EMA product number, Medicine status, Opinion status**, Active substance, Species (vet.)**, ATC code, Biosimilar, Generic, Orphan medicine, Opinion adopted date**, Revision number**, …
DPD	Health Can.	TXT/CSV	42,042	41,636	87	DIN, brand name, ingredient, number of AIs, strength, dosage unit**, company name, class, ATC code, schedule, status, status date, form, route, descriptor**, …

*Source: Downloaded directly from official provider websites. **Entities:** Distinct normalized brand-name and active-substance pairs.
**Missingness: ** indicates >30% missing values (mostly structural, such as optional notes, veterinary fields, or specific regulatory events).

Table 1 summarises the datasets and their most relevant attributes, including those with high missingness.

Drugs@FDA

Drugs@FDA contains 51,839 product rows across 29,377 applications, forming 8,175 normalized medicine entities. Among applications, 78.3% are ANDAs, 20.1% NDAs, and 1.7% BLAs. Three attributes exceed 30% missingness: `ApplPublicNotes` (99.99%), `SubmissionsPublicNotes` (99.83%), and `ReviewPriority` (41.73%).

EMA Medicines

EMA Medicines contains 2,746 medicine rows, forming 2,620 normalized medicine entities, with no duplicate input rows. Human medicines account for 85.6% of the records. By status, 68.5% are authorised, 15.4% withdrawn, and 10.4% are withdrawn applications. Twelve attributes exceed 30% missingness, mostly because veterinary fields do not apply to human medicines and regulatory-event dates apply only to particular statuses.

Health Canada Drug Product Database (DPD)

The Health Canada DPD contains 58,137 product records across 12 tables linked by `DRUG_CODE`. They are consolidated into 42,042 medicine rows, forming 41,636 normalized medicine entities. Human products account for 87.7% of the records. By status, 66.5% are cancelled and 23.0% marketed. Notable missingness includes ATC code (21.47%) and first marketed date (11.67%).

2.2 Integrated schema and overlap with input schemata

The integrated schema contains 16 attributes (Table 2); eleven occur in at least two sources. `activeSubstances` and `strengths` are list-valued, while the remaining 14 attributes are single-valued (including the numeric attribute `activeSubstanceCount`). Strengths are represented as numeric value–unit pairs.

2.3 Entity Identification and Count Requirement

A medicine is identified by its `brandName` and `activeSubstances`. Contrary to initial assumptions, `licenceHolder` cannot be used as an identifying attribute because the same medicine is frequently marketed by different regional subsidiaries or unrelated licensing partners across jurisdictions (for example, *Verkazia* is held by `HARROW EYE` in the US, `Santen Oy` in the EU, and `APOTEX INC` in Canada).

To estimate how many medicines appear in multiple datasets based on exact matching, we build a normalized key (`medicine_id`) from `brandName` and `activeSubstances` by converting text to upper case, removing accents and dosage strengths, standardizing punctuation, and sorting multiple active substances. Let F, E and D denote the sets of `medicine_id` values in Drugs@FDA, EMA Medicines and the DPD. The pairwise exact overlaps are $|F \cap E| = 428$, $|F \cap D| = 949$ and $|E \cap D| = 594$, yielding a deduplicated union of $|(F \cap E) \cup (F \cap D) \cup (E \cap D)| = 1{,}407$ distinct medicines – well above the required minimum of 1,000.

Table 2: Attribute Intersection with Integrated Schema

Attribute name	**Attribute type**	**Datasets in which the attribute is found**
`brandName`	string	Drugs@FDA, EMA Medicines, DPD
`activeSubstances`	list (string)	Drugs@FDA, EMA Medicines, DPD
`activeSubstanceCount`	numeric (integer)	Drugs@FDA (derived), EMA Medicines (derived), DPD
`licenceHolder`	string	Drugs@FDA, EMA Medicines, DPD
`regulatoryStatus`	string (categorical)	Drugs@FDA, EMA Medicines, DPD
`productClass`	string (categorical)	Drugs@FDA (derived), EMA Medicines, DPD
`atcCode`	string	EMA Medicines, DPD
`generic`	boolean	Drugs@FDA (derived), EMA Medicines
`biosimilar`	boolean	EMA Medicines
`orphanMedicine`	boolean	EMA Medicines
`dosageForm`	string	Drugs@FDA, DPD
`route`	string	Drugs@FDA, DPD
`strengths`	list (numeric value, unit)	Drugs@FDA, DPD
`usApprovalDate`	date	Drugs@FDA
`euAuthorisationDate`	date	EMA Medicines
`caStatusDate`	date	DPD

However, exact matching misses many identical medicines because both identifying attributes can vary across countries (Table 3):

- **Active substance naming and salt forms:** The same substance can have different regional spellings (e.g., `CYCLOSPORINE` in the US and Canada vs. `ciclosporin` in the EU for *Verkazia*) or different chemical and salt representations (e.g., `GLYCOPYRROLATE` in the US, `glycopyrronium bromide` in the EU, and `GLYCOPYRRONIUM (GLYCOPYRRONIUM BROMIDE)` in Canada for *Seebri*).
- **Brand name suffixes:** The same medicine may carry different device or formulation suffixes across markets, such as `SEEBRI NEOHALER` in Drugs@FDA versus `Seebri Breezhaler` in EMA Medicines and the DPD.

Table 3: Examples of identical medicines with differing identifying attributes across sources

Dataset	**Brand Name**	**Active Substance**
Drugs@FDA	`VERKAZIA`	`CYCLOSPORINE`
EMA Medicines	`Verkazia`	`ciclosporin`
Health Canada DPD	`VERKAZIA`	`CYCLOSPORINE`
Drugs@FDA	`SEEBRI NEOHALER`	`GLYCOPYRROLATE`
EMA Medicines	`Seebri Breezhaler`	`glycopyrronium bromide`
Health Canada DPD	`SEEBRI BREEZHALER`	`GLYCOPYRRONIUM (GLYCOPYRRONIUM BROMIDE)`